
Representation Regularization and Theoretical Mechanics of Joint-Embedding Predictive Architectures for Multi-Modal Brain MRI: A Comprehensive Analysis of Label Efficiency, Non-Collapse Dynamics, and Out-of-Distribution Generalization

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Abstract

Joint-Embedding Predictive Architectures (JEPA) represent a paradigm shift in self-supervised visual representation learning, operating entirely in latent embedding space without pixel-level image reconstruction. Standard I-JEPA prevents latent representation collapse through an Exponential Moving Average (EMA) momentum update of a teacher target encoder. Recent theoretical formulations—specifically SIGReg (Sketched Isotropic Gaussian Regularization in LeJEPA) and VISReg (Variance-Invariance-Sketching Regularization)—introduce analytical feature regularization penalties that enable *heuristic-free* single-encoder JEPA training without momentum teachers. In this paper, we present a mathematically rigorous and comprehensive evaluation of I-JEPA, SigReg JEPA, VisReg JEPA, a 2D Residual UNet baseline, and a state-of-the-art 2D nnU-Net baseline with deep supervision on 2D multi-modal Glioma MRI (BraTS GLI). We detail the mathematical formulations and architectural micro-designs across all models, including internal residual skip connections, multi-scale encoder-to-decoder concatenation skip connections, pure bottleneck transposed convolutional decoding, stochastic patch block masking with batch-uniform collation, Random Modality Dropout with guaranteed active channel fallbacks, hyperspherical projection sketching, and gradient-clipped optimization schedules. We evaluate latent collapse metrics (Effective Feature Rank Rank_{eff} , Average Pairwise Cosine Similarity S_{cosine}), downstream segmentation metrics (Dice Similarity Coefficient, Intersection over Union, 95th Percentile Hausdorff Distance HD95), and out-of-distribution transfer across three clinical regimes: (1) Full-Data Benchmark, (2) Low-Data Label Efficiency (1% to 100% labels), and (3) Out-of-Distribution (OOD) Scanner Shift and Zero-Shot Cross-Pathology Transfer on Meningioma MRI (BraTS-MEN-RT). Our empirical findings establish that under extreme label scarcity (1% labels / 63 training slices), SigReg JEPA achieves a top downstream Test Dice of 0.5933, outperforming fully supervised nnU-Net (0.3847) by +20.86% absolute Dice (+54.2% relative improvement) while executing fine-tuning $1.9\times$ faster per epoch (0.75 s/epoch vs 1.42 s/epoch). Furthermore, under zero-padded missing modality shifts, self-supervised JEPAs preserve zero-shot transferability (0.0700–0.0929 Dice) whereas standard UNet completely collapses (0.0151 Dice), proving that analytical variance-covariance regularization builds

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modality-invariant, high-rank representations while eliminating momentum teacher overhead.

1 Introduction and Theoretical Framework

Automated segmentation of brain gliomas from multi-modal Magnetic Resonance Imaging (MRI)—comprising T1-weighted (T1), post-contrast T1-weighted (T1c), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) sequences—is essential for clinical neuro-oncology diagnosis, surgical boundary planning, and longitudinal monitoring [Baid et al., 2024]. Supervised deep convolutional architectures, such as Residual UNet [Ronneberger et al., 2015] and nnU-Net [Isensee et al., 2021], achieve high accuracy when extensive pixel-wise ground-truth annotations are available. However, expert manual voxel delineation is labor-intensive, scarce, and susceptible to inter-observer variability.

Self-Supervised Learning (SSL) addresses this bottleneck by pre-training representation encoders on unannotated MRI volumes prior to downstream task fine-tuning. Unlike Masked Autoencoders (MAE) [He et al., 2022] that reconstruct high-frequency pixel intensities in pixel space, Joint-Embedding Predictive Architectures (JEPA) [Assran et al., 2023] operate exclusively in latent embedding space. JEPA models predict target patch representations from context patch representations, forcing the encoder to learn abstract semantic structures rather than low-level noise.

A central challenge in energy-based self-supervised architectures is preventing *latent representation collapse*—a trivial state where the encoder maps all input patches to a constant vector $z = c \in \mathbb{R}^D$, yielding zero prediction error $\mathcal{L} = 0$ without learning meaningful semantics.

Standard I-JEPA [Assran et al., 2023] prevents collapse through an Exponential Moving Average (EMA) momentum update of a target teacher encoder $E_{\bar{\theta}}$. However, EMA updating is a heuristic mechanism that introduces computational memory overhead, requires hyperparameter tuning of the momentum parameter m , and lacks formal non-collapse guarantees.

Recent theoretical formulations derived from statistical distribution matching and sketching principles, specifically SIGReg [Balestriero and LeCun, 2025] and VISReg [Wu et al., 2026], introduce analytical feature regularization penalties applied directly to online encoder activations. By constraining representations to match isotropic Gaussian distributions via 1D random projections, these methods establish a *heuristic-free* single-encoder JEPA framework that eliminates EMA target teachers.

In this paper, we provide a complete mathematical and empirical breakdown of JEPA architectures for multi-modal brain MRI segmentation. We formalize loss objectives, backpropagation decoupling, training protocols, probing metrics, and evaluation procedures across full-data, low-data, and out-of-distribution transfer settings.

2 Mathematical Methodology and Architecture Formulations

2.1 Multi-Modal Input Processing and Random Modality Dropout

Let $X \in \mathbb{R}^{B \times 4 \times H \times W}$ represent a batch of 2D axial MRI slices with spatial dimensions $H = W = 240$ and $C = 4$ co-registered sequence channels: $X = [X_{T1}, X_{T1c}, X_{T2}, X_{FLAIR}]$.

Physical Sequence Characteristics & Clinical Tumor Pathophysiology Glioma diagnosis and boundary delineation require multi-parametric MRI because different pulse sequences probe orthogonal biological tissue properties [Baid et al., 2024]:

- **T1-weighted (T1):** Characterized by short repetition time (TR) and echo time (TE), T1 reflects longitudinal spin-lattice relaxation (T_1). It yields sharp contrast between healthy gray matter, white matter, and cerebrospinal fluid (CSF), providing anatomical reference geometry.
- **Contrast-Enhanced T1-weighted (T1c):** Administered via intravenous Gadolinium-based contrast agents (e.g., Gd-DTPA). Paramagnetic gadolinium ions accelerate local water proton T_1 relaxation, producing intense signal enhancement strictly where the blood-brain

barrier (BBB) is structurally disrupted by tumor neovascularization. T1c delineates the actively proliferating, viable angiogenic tumor boundary.

- **T2-weighted (T2):** Utilizing long TR and TE, T2 probes transverse spin-spin relaxation (T_2). It is highly sensitive to free and bound water accumulation, clearly depicting diffuse hyperintense vasogenic edema and bulk mass effect.
- **Fluid-Attenuated Inversion Recovery (FLAIR):** Applies an initial 180° inversion pulse with an inversion time $TI \approx 2000$ ms precisely timed to the null point of free water. By suppressing the bright signal of ventricular CSF, FLAIR isolates non-enhancing peritumoral infiltrative edema and satellite tumor margins that would otherwise be masked by adjacent ventricles.

Prior to slice extraction, non-zero brain tissue voxels are Z-score normalized independently per 3D volume sequence:

$$\hat{X}_c(v) = \frac{X_c(v) - \mu_c}{\sigma_c + \epsilon}, \quad \forall c \in \{T1, T1c, T2, FLAIR\} \quad (1)$$

where μ_c and σ_c denote the empirical mean and standard deviation computed strictly over non-zero intracranial voxels in volume sequence c , and $\epsilon = 10^{-8}$ ensures numerical stability.

Random Modality Dropout (RMD) with Guaranteed Non-Empty Fallback In routine clinical oncology, complete 4-sequence acquisitions are frequently compromised by emergency patient agitation, metallic implants, acquisition time constraints, or severe renal impairment contraindicating gadolinium injection (nephrogenic systemic fibrosis) [Havaei et al., 2017, Dorent et al., 2019]. Models naively trained on complete 4-channel tensors develop brittle inter-channel co-dependencies and collapse when modalities are omitted.

To instill intrinsic modality invariance during training, we propose **Random Modality Dropout (RMD)** with a guaranteed non-empty fallback mechanism, formalized in Algorithm 1.

Algorithm 1 Random Modality Dropout with Guaranteed Non-Empty Fallback

Require: Normalized multi-channel slice batch $\hat{X} \in \mathbb{R}^{B \times C \times H \times W}$, dropout probability $p_{\text{drop}} \in [0, 1)$, training flag $\tau \in \{0, 1\}$.

Ensure: Modality-augmented batch $\tilde{X} \in \mathbb{R}^{B \times C \times H \times W}$.

- 1: **if** $\tau = 0$ **or** $p_{\text{drop}} \leq 0$ **then**
 - 2: **return** $\hat{X}$ ▷ Deactivated during inference and validation
 - 3: **end if**
 - 4: Sample channel mask: $\mathbf{M} \leftarrow \mathbb{I}(\mathbf{U} > p_{\text{drop}})$ where $\mathbf{U} \sim \mathcal{U}(0, 1)^{B \times C \times 1 \times 1}$
 - 5: Identify completely zeroed slices: $\mathbf{A}_{\text{zero}} \leftarrow \left(\sum_{c=1}^C \mathbf{M}_{:,c,1,1} == 0 \right) \in \{0, 1\}^B$
 - 6: Sample fallback channels: $c^* \sim \text{Uniform}(\{1, \dots, C\})^B$
 - 7: Construct fallback one-hot mask: $\mathbf{M}_{\text{fallback}} \leftarrow \text{OneHot}(c^*) \in \mathbb{R}^{B \times C \times 1 \times 1}$
 - 8: Reconcile mask: $\mathbf{M} \leftarrow \text{where}(\mathbf{A}_{\text{zero}}, \mathbf{M}_{\text{fallback}}, \mathbf{M})$
 - 9: **return** $\tilde{X} = \hat{X} \odot \mathbf{M}$ ▷ Element-wise broadcast multiplication
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Spatial Data Augmentation Randomization During downstream fine-tuning, training slices undergo stochastic spatial transformations:

- **Random Horizontal Flip:** Spatial axis 0 inverted with probability $p = 0.5$.
- **Random Vertical Flip:** Spatial axis 1 inverted with probability $p = 0.5$.
- **Random Affine Rotation:** Uniform spatial rotation angle $\theta \sim \mathcal{U}(-17.2^\circ, +17.2^\circ)$ (range 0.3 rad) with probability $p = 0.5$, applying bilinear interpolation to image intensities and nearest-neighbor interpolation to discrete segmentation ground-truth.

2.2 Vision Transformer Backbone Architecture & Internal Skip Connections

The vision encoder E_θ processes 2D multi-modal slices using non-overlapping $P \times P = 16 \times 16$ patch embeddings. A 2D slice with $H = W = 240$ is partitioned into a spatial grid of patch tokens:

$$N = \left(\frac{H}{P}\right) \times \left(\frac{W}{P}\right) = \left(\frac{240}{16}\right) \times \left(\frac{240}{16}\right) = 15 \times 15 = 225 \text{ patches} \quad (2)$$

A 2D convolution with kernel size 16, stride 16, and 4 input channels maps each $16 \times 16 \times 4$ patch to embedding dimension $D = 384$. Learnable 1D spatial positional embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1 \times 225 \times 384}$, initialized from a truncated normal distribution $\mathcal{TN}(0, 0.02^2)$, are added:

$$\mathbf{Z}^{(0)} = \text{Conv2d}_{16 \times 16}(\tilde{X}) + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{B \times 225 \times 384} \quad (3)$$

The patch tokens pass through $L = 8$ Transformer Encoder blocks. Each block employs a **Pre-Layer Normalization (Pre-LN)** design featuring internal residual skip connections around both the attention and feed-forward sub-layers:

$$\mathbf{Z}'^{(l)} = \mathbf{Z}^{(l-1)} + \text{MHSA}\left(\text{LN}(\mathbf{Z}^{(l-1)})\right) \quad (4)$$

$$\mathbf{Z}^{(l)} = \mathbf{Z}'^{(l)} + \text{MLP}\left(\text{LN}(\mathbf{Z}'^{(l)})\right), \quad l = 1, \dots, 8 \quad (5)$$

The residual shortcut additions in Equations (4) and (5) preserve identity gradient highways throughout the depth of the encoder, preventing gradient attenuation. The Multi-Head Self-Attention (MHSA) module operates across $h = 6$ attention heads (head dimension $d_k = D/h = 64$):

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

The MLP sub-layer consists of a two-layer feed-forward network with expansion ratio 4 ($D_{\text{ff}} = 1536$) and GELU non-linearities: $\text{MLP}(x) = \text{Linear}_{1536 \rightarrow 384}(\text{GELU}(\text{Linear}_{384 \rightarrow 1536}(x)))$.

Vectorized Subset Patch Encoding When encoding context patches during self-supervised pre-training, only context indices $\mathcal{I}_{\text{ctx}} \subset \{1, \dots, N\}$ ($|\mathcal{I}_{\text{ctx}}| = N_{\text{ctx}}$) are passed to the encoder. To completely prevent attention leakage across masked target patches without inefficient $O(B)$ Python loops, we vectorize subset selection using `torch.gather`:

$$\mathbf{Z}_{\text{ctx}}^{(0)} = \text{gather}\left(\mathbf{Z}^{(0)}, \text{dim} = 1, \text{index} = \mathcal{I}_{\text{ctx}}\right) \in \mathbb{R}^{B \times N_{\text{ctx}} \times 384} \quad (7)$$

Self-attention is then computed strictly over context tokens, guaranteeing zero computational cost and zero information leakage from masked target regions.

2.3 Stochastic Patch Masking: JEPAMaskingTransform

The self-supervised prediction task requires partitioning the 15×15 grid into context and target patch sets. We implement a stochastic block masking transform (JEPAMaskingTransform) designed to satisfy three criteria: (1) realistic spatial context, (2) zero overlap between context and targets, and (3) uniform tensor dimensions for dense batch collation.

1. **Target Block Sampling:** We sample $K = 4$ target blocks. Each target block has a fixed spatial geometry of $H_{\text{tgt}} \times W_{\text{tgt}} = 5 \times 5 = 25$ patches ($N_{\text{tgt}} = 25$). The top-left corner coordinates (t_k, l_k) for each block $k \in \{1, \dots, K\}$ are sampled uniformly:

$$t_k \sim \mathcal{U}\{0, H_{\text{grid}} - H_{\text{tgt}}\}, \quad l_k \sim \mathcal{U}\{0, W_{\text{grid}} - W_{\text{tgt}}\} \quad (8)$$

The target patch indices for block k are $\mathcal{I}_{\text{tgt},k} = \{(t_k + r)W_{\text{grid}} + (l_k + c) \mid 0 \leq r < 5, 0 \leq c < 5\}$. The composite target set is $\mathcal{I}_{\text{tgt}} = \bigcup_{k=1}^K \mathcal{I}_{\text{tgt},k}$.

2. **Candidate Context Window:** A large candidate context block of spatial dimension $H_{\text{ctx}} \times W_{\text{ctx}} = 14 \times 14 = 196$ patches is sampled from the grid:

$$t_{\text{ctx}} \sim \mathcal{U}\{0, H_{\text{grid}} - H_{\text{ctx}}\}, \quad l_{\text{ctx}} \sim \mathcal{U}\{0, W_{\text{grid}} - W_{\text{ctx}}\} \quad (9)$$

3. **Overlap Filtering:** Any candidate patch overlapping with any target block is strictly pruned:

$$\mathcal{C}_{\text{clean}} = \mathcal{I}_{\text{ctx, cand}} \setminus \mathcal{I}_{\text{tgt}} \quad (10)$$

4. **Uniform Batch Collation Replenishment:** Dense GPU tensor collation requires every sample in batch B to have identical context length $N_{\text{ctx}} = 96$ patches. If $|\mathcal{C}_{\text{clean}}| \geq N_{\text{ctx}}$, we select the first N_{ctx} patches. If $|\mathcal{C}_{\text{clean}}| < N_{\text{ctx}}$, the deficit is replenished by drawing uniformly at random from the non-target grid pool $\mathcal{P}_{\text{non-tgt}} = \{1, \dots, N\} \setminus \mathcal{I}_{\text{tgt}}$:

$$\mathcal{I}_{\text{ctx}} = \mathcal{C}_{\text{clean}} \cup \text{Sample}(\mathcal{P}_{\text{non-tgt}} \setminus \mathcal{C}_{\text{clean}}, N_{\text{ctx}} - |\mathcal{C}_{\text{clean}}|) \quad (11)$$

This guarantees $|\mathcal{I}_{\text{ctx}}| = 96$ for every slice, eliminating variable-length padding and ragged tensors.

2.4 JEPA Predictor Architecture & Decoupled Vectorized Gather

The predictor P_ϕ is a narrow 4-block Transformer operating in a lower-dimensional projection space $D_{\text{pred}} = 192$ (half the encoder embedding dimension $D = 384$).

Dimension Projection & Mask Token Context tokens are projected via a linear layer: $\mathbf{Z}_{\text{ctx, proj}} = \text{Linear}_{384 \rightarrow 192}(\mathbf{Z}_{\text{ctx}}) \in \mathbb{R}^{B \times N_{\text{ctx}} \times 192}$. A shared learnable mask token $\mathbf{e}_{\text{mask}} \in \mathbb{R}^{1 \times 1 \times 192}$, initialized via truncated normal $\mathcal{TN}(0, 0.02^2)$, serves as the placeholder for target patches to be predicted.

Vectorized Positional Embedding Retrieval Because context indices $\mathcal{I}_{\text{ctx}}$ and target indices $\mathcal{I}_{\text{tgt}}$ can independently be 1D (shared across the batch) or 2D (heterogeneous per-sample masks), we decouple positional retrieval using `torch.gather`:

$$\mathbf{P}_{\text{ctx}} = \begin{cases} \mathbf{E}_{\text{pos, pred}}[:, \mathcal{I}_{\text{ctx}}, :].\text{expand}(B, -1, -1), & \text{if } \dim(\mathcal{I}_{\text{ctx}}) = 1 \\ \text{gather}(\mathbf{E}_{\text{pos, pred}}.\text{expand}(B, -1, -1), 1, \mathcal{I}_{\text{ctx}} \otimes \mathbf{1}_{D_{\text{pred}}}), & \text{if } \dim(\mathcal{I}_{\text{ctx}}) = 2 \end{cases} \quad (12)$$

and similarly for target positional embeddings $\mathbf{P}_{\text{tgt}}$ using $\mathcal{I}_{\text{tgt}}$.

Predictor Forward Pass We construct the target mask token sequence as $\mathbf{M}_{\text{tgt}} = \mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}$ expanded across batch and target dimensions $(B, N_{\text{tgt}}, D_{\text{pred}})$. The context and target tokens are concatenated along the sequence dimension:

$$\mathbf{Z}_{\text{pred}}^{(0)} = [(\mathbf{Z}_{\text{ctx, proj}} + \mathbf{P}_{\text{ctx}}); \mathbf{M}_{\text{tgt}}] \in \mathbb{R}^{B \times (N_{\text{ctx}} + N_{\text{tgt}}) \times 192} \quad (13)$$

$\mathbf{Z}_{\text{pred}}^{(0)}$ passes through 4 Pre-LN Transformer blocks with residual skip connections ($h = 6$, $D_{\text{ff}} = 768$). The target positions are extracted from the sequence and projected back to the encoder dimension:

$$\hat{Y}_{\text{tgt}} = \text{Linear}_{192 \rightarrow 384} \left(\text{LN} \left(P_\phi(\mathbf{Z}_{\text{pred}}^{(0)})[:, N_{\text{ctx}} :, :] \right) \right) \in \mathbb{R}^{B \times N_{\text{tgt}} \times 384} \quad (14)$$

2.5 I-JEPA Mathematical Formulation (Dual-Encoder with EMA Teacher)

Standard I-JEPA [Assran et al., 2023] maintains two encoders: an online context encoder E_θ and a target teacher encoder $E_{\bar{\theta}}$. Target representations are computed by passing the full unmasked slice through $E_{\bar{\theta}}$:

$$Y_{\text{tgt}} = E_{\bar{\theta}}(X)[:, \mathbf{I}_{\text{tgt}}, :] \in \mathbb{R}^{B \times N_{\text{tgt}} \times 384} \quad (15)$$

Asymmetric Target Normalization Following the official I-JEPA formulation [Assran et al., 2023], a crucial architectural design choice is that **only target tokens are Layer-Normalized**, whereas predicted tokens $\hat{Y}_{\text{tgt}}$ remain unnormalized:

$$\tilde{Y}_{\text{tgt}} = \text{LayerNorm}(Y_{\text{tgt}}) = \frac{Y_{\text{tgt}} - \mu(Y_{\text{tgt}})}{\sqrt{\sigma^2(Y_{\text{tgt}}) + \epsilon}} \odot \gamma + \beta \quad (16)$$

If predictions $\hat{Y}_{\text{tgt}}$ were also normalized, the predictor could artificially alter its internal weight magnitudes without incurring gradient penalties, causing representational instability. Normalizing the teacher targets establishes a standardized, zero-mean, unit-variance coordinate reference for the online network.

Prediction Objective and Huber Loss Robustness The prediction loss is computed using Smooth L_1 (Huber) loss between predicted tokens $\hat{Y}_{\text{tgt}}$ and normalized teacher tokens $\tilde{Y}_{\text{tgt}}$:

$$\mathcal{L}_{\text{I-JEPA}}(\theta, \phi) = \frac{1}{K} \sum_{k=1}^K \frac{1}{B \cdot N_{\text{tgt}} \cdot D} \sum_{b=1}^B \sum_{i=1}^{N_{\text{tgt}}} \sum_{j=1}^D \text{Smooth}_{L_1} \left(\hat{Y}_{b,i,j}^{(k)} - \tilde{Y}_{b,i,j}^{(k)} \right) \quad (17)$$

where Smooth L_1 transitions between quadratic and linear penalties:

$$\text{Smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1.0 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (18)$$

In multi-modal brain MRI, hyperintense necrotic cores, contrast-enhancing margins, or skull-stripping boundary artifacts produce heavy-tailed representation residuals during early pre-training epochs. Standard mean squared error (MSE, L_2) produces quadratic gradient penalties that explode on outliers, whereas Smooth L_1 bounds the gradient magnitude to ± 1 , stabilizing optimization.

Teacher Non-Differentiability & Evaluation Mode Enforcement The target teacher encoder parameters $\bar{\theta}$ do not receive gradients ($\nabla_{\bar{\theta}} \mathcal{L} = 0$). Instead, they are updated at training step t via Exponential Moving Average (EMA) momentum $m \in [0.996, 1.0]$ [Grill et al., 2020, Assran et al., 2023]:

$$\bar{\theta}_t \leftarrow m\bar{\theta}_{t-1} + (1 - m)\theta_t \quad (19)$$

Crucially, during training, the teacher encoder must strictly remain in evaluation mode:

$$E_{\bar{\theta}}.\text{eval}() \equiv \text{True}, \quad \forall \text{ training iterations } t \quad (20)$$

This overrides the global PyTorch model training state to guarantee that dropout layers in $E_{\bar{\theta}}$ are deactivated and running statistics remain stationary, ensuring completely deterministic target generation.

2.6 SigReg JEPA Mathematical Formulation (Heuristic-Free Single-Encoder)

SigReg JEPA (LeJEPA) [Balestrieri and LeCun, 2025] completely eliminates the EMA teacher target encoder $E_{\bar{\theta}}$, executing self-supervised pre-training using a single encoder E_{θ} . Target representations are generated directly by the online encoder E_{θ} under `torch.no_grad()` without momentum copies.

Cramér-Wold Theorem and Projector MLP To mathematically prevent representation collapse without an EMA teacher, Sketched Isotropic Gaussian Regularization (SIGReg) enforces that the empirical distribution of representations matches a standard isotropic Gaussian $\mathcal{N}(0, \mathbf{I}_{d_p})$ [Balestrieri and LeCun, 2025].

By the **Cramér-Wold theorem** [Cramér and Wold, 1936], a multivariate probability measure on $\mathbb{R}^{d_p}$ is uniquely determined by the family of its 1D marginal distributions under all 1D linear projections onto the unit hypersphere $\mathbb{S}^{d_p-1}$. To isolate this regularization from downstream spatial features, representations pass through a 2-layer Projector MLP $g_{\psi} : \mathbb{R}^D \rightarrow \mathbb{R}^{d_p}$ ($d_p = 128$):

$$Z_{\text{proj}} = g_{\psi}(Z_{\text{ctx}}) = \text{Linear}_{1024 \rightarrow 128}(\text{GELU}(\text{LayerNorm}(\text{Linear}_{384 \rightarrow 1024}(Z_{\text{ctx}})))) \quad (21)$$

Epps-Pulley Test Statistic & Quadrature We sample $M = 256$ random projection directions on the unit hypersphere: $u_m = v/\|v\|_2$, where $v \sim \mathcal{N}(0, \mathbf{I}_{d_p})$. The 1D projections for $N = B \cdot N_{\text{ctx}}$ tokens are $p_m = Z_{\text{proj}}u_m \in \mathbb{R}^N$.

The **Epps-Pulley test statistic** [Epps and Pulley, 1983] compares the empirical characteristic function (ECF) $\hat{\phi}_N(t) = \frac{1}{N} \sum_{n=1}^N \exp(itp_n)$ to the analytical standard Gaussian characteristic function $\phi_0(t) = \exp(-t^2/2)$ weighted by the standard normal density $d\mu(t) = \exp(-t^2/2)dt$:

$$\mathcal{T}_{\text{EP}}(p) = N \int_{-\infty}^{\infty} |\hat{\phi}_N(t) - \phi_0(t)|^2 d\mu(t) \quad (22)$$

Exploiting symmetry across $t \in [-t_{\text{max}}, t_{\text{max}}]$, the integral is evaluated over $[0, t_{\text{max}}]$ ($t_{\text{max}} = 3.0$) via trapezoidal quadrature with $Q = 17$ knot points and step size $\Delta t = t_{\text{max}}/(Q - 1)$:

$$w_0 = w_{Q-1} = \Delta t, \quad w_k = 2\Delta t \quad (k = 1, \dots, Q - 2) \quad (23)$$

The discretized statistic evaluates to:

$$\mathcal{T}_{\text{EP}}(p) = N \sum_{k=0}^{Q-1} w_k e^{-t_k^2/2} \left[\left(\frac{1}{N} \sum_{n=1}^N \cos(t_k p_n) - e^{-t_k^2/2} \right)^2 + \left(\frac{1}{N} \sum_{n=1}^N \sin(t_k p_n) \right)^2 \right] \quad (24)$$

Mathematical Derivation of the Sample Count Scaling Factor N A critical implementation subtlety in Epps–Pulley regularization involves the scaling factor N :

$$\frac{\partial \hat{\phi}_N(t)}{\partial p_i} = \frac{\partial}{\partial p_i} \left(\frac{1}{N} \sum_{n=1}^N \exp(itp_n) \right) = \frac{it}{N} \exp(itp_i) \quad (25)$$

Because the derivative of the sample average ECF contains an intrinsic $\frac{1}{N}$ factor, the gradient of the squared error integral without N scales as $\mathcal{O}(1/N)$. In mini-batch pre-training with $N = B \cdot N_{\text{ctx}} = 8 \times 96 = 768$ tokens, omitting N dilutes the regularization gradient by $1/768$, causing representation gradients to vanish ($\|\nabla_z \mathcal{L}\| \approx 0.00019$). Multiplying by N cancels the $\frac{1}{N}$ derivative denominator:

$$\nabla_{p_i} \left[N \cdot |\hat{\phi}_N(t) - \phi_0(t)|^2 \right] = 2 \cdot \text{Re} \left\{ (\hat{\phi}_N(t) - \phi_0(t)) \cdot it \exp(-itp_i) \right\} = \mathcal{O}(1) \quad (26)$$

This restores the gradient norm to $\|\nabla_z \mathcal{L}\| \approx 0.1294$, matching the asymptotic χ^2 distribution under H_0 and achieving perfect equilibrium with the JEPA prediction loss. The complete pre-training workflow for SigReg JEPA is formalized in Algorithm 2.

Algorithm 2 Self-Supervised Pre-Training of SigReg JEPA (LeJEPA)

Require: 4-channel MRI slice batch $X \in \mathbb{R}^{B \times 4 \times 240 \times 240}$, Encoder E_θ , Predictor P_ϕ , Projector g_ψ , Masking transform `JEPAMaskingTransform`, projection count $M = 256$, regularizer weight $\lambda_{\text{SIGReg}} = 1.0$.

Ensure: Updated parameter sets θ, ϕ, ψ .

- 1: Mask image: $\{\mathcal{I}_{\text{ctx}}, \{\mathcal{I}_{\text{tgt}}^{(k)}\}_{k=1}^K\} \leftarrow \text{JEPAMaskingTransform}(X)$
 - 2: Forward online encoder on full image without gradients:
 - 3: $Y_{\text{full}} \leftarrow E_\theta(X).detach() \in \mathbb{R}^{B \times 225 \times D}$ ▷ Target representations
 - 4: Forward online encoder on context patches:
 - 5: $Z_{\text{ctx}} \leftarrow E_\theta(X, \text{patch_indices} = \mathcal{I}_{\text{ctx}}) \in \mathbb{R}^{B \times N_{\text{ctx}} \times D}$
 - 6: Predict target representations:
 - 7: $\hat{Y}_{\text{tgt}}^{(k)} \leftarrow P_\phi(Z_{\text{ctx}}, \mathcal{I}_{\text{ctx}}, \mathcal{I}_{\text{tgt}}^{(k)})$ for $k = 1, \dots, K$
 - 8: Compute prediction loss: $\mathcal{L}_{\text{pred}} \leftarrow \frac{1}{K} \sum_{k=1}^K \text{Smooth}_{L_1} \left(\hat{Y}_{\text{tgt}}^{(k)}, \text{LayerNorm}(Y_{\text{full}}[:, \mathcal{I}_{\text{tgt}}^{(k)}, :]) \right)$
 - 9: Project context tokens: $Z_{\text{proj}} \leftarrow g_\psi(Z_{\text{ctx}}) \in \mathbb{R}^{(B \cdot N_{\text{ctx}}) \times d_p}$
 - 10: Sample M random unit projection vectors $A \sim \mathcal{N}(0, \mathbf{I}_{d_p \times M})$, $A \leftarrow \text{Normalize}(A, \text{dim} = 0)$
 - 11: 1D random projections: $P \leftarrow Z_{\text{proj}} A \in \mathbb{R}^{N \times M}$
 - 12: Compute Epps–Pulley test statistic: $\mathcal{L}_{\text{SIGReg}} \leftarrow \frac{1}{M} \sum_{m=1}^M \mathcal{T}_{\text{EP}}(P[:, m])$
 - 13: Total loss: $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}} \mathcal{L}_{\text{SIGReg}}$
 - 14: Backpropagate and update: $\theta, \phi, \psi \leftarrow \text{AdamW}(\nabla_{\theta, \phi, \psi} \mathcal{L}_{\text{total}})$
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2.7 VisReg JEPA: Projector MLP & Standardized Sliced-Wasserstein Decoupling

VisReg JEPA [Wu et al., 2026] similarly operates without an EMA teacher encoder. It enforces representation non-collapse by decoupling scale regularization from distribution shape regularization via Optimal Transport [Bonneel et al., 2015].

Dedicated Projector MLP To isolate non-collapse regularization from downstream feature representations, VisReg JEPA routes context tokens through a 2-layer Projector MLP g_ψ :

$$Z_{\text{proj}} = \text{Linear}_{1024 \rightarrow 128} (\text{GELU} (\text{LayerNorm} (\text{Linear}_{384 \rightarrow 1024} (Z_{\text{ctx}})))) \in \mathbb{R}^{B \times N_{\text{ctx}} \times 128} \quad (27)$$

The raw backbone features Z_{ctx} remain spatially expressive and localized for downstream decoding, while the projected features Z_{proj} satisfy isotropic Gaussianity.

Scale Regularization (Batch Variance Hinge) Scale regularization forces feature variance across the batch along each projected dimension to meet or exceed a target standard deviation $\sigma_{\text{target}} = 1.0$:

$$\mathcal{L}_{\text{var}}(Z_{\text{proj}}) = \frac{1}{d_p} \sum_{j=1}^{d_p} \max \left(0, 1.0 - \sqrt{\text{Var}(Z_{\text{proj}, :, j})} + \epsilon \right) \quad (28)$$

When feature variance is at or above 1.0, this hinge loss evaluates to identically zero, allowing representations to expand without penalty.

Shape Regularization via Standardized Projections $M = 256$ random projection vectors u_m are sampled uniformly from the unit hypersphere: $u \sim \mathcal{N}(0, \mathbf{I}_{d_p})$, $u_m = u/\|u\|_2 \in \mathbb{S}^{d_p-1}$. 1D slices are formed by matrix multiplication: $p_m = Z_{\text{proj}} u_m \in \mathbb{R}^N$ ($N = B \cdot N_{\text{ctx}}$).

To strictly isolate the *shape* of the distribution from variance and mean shifts, we standardize each 1D projected slice prior to quantile sorting:

$$\tilde{p}_{im} = \frac{p_{im} - \mu_m}{\sigma_m + \epsilon}, \quad \mu_m = \frac{1}{N} \sum_{i=1}^N p_{im}, \quad \sigma_m = \sqrt{\frac{1}{N} \sum_{i=1}^N (p_{im} - \mu_m)^2 + \epsilon} \quad (29)$$

Sorting the standardized projections $\tilde{p}_{(\cdot)m}$ yields empirical quantiles. The Sliced-Wasserstein Distance against standard normal $\mathcal{N}(0, 1)$ quantiles has an exact closed-form solution:

$$\mathcal{L}_{\text{SWD}}(Z_{\text{proj}}) = \frac{1}{M} \sum_{m=1}^M \frac{1}{N} \sum_{i=1}^N \left| \tilde{p}_{(i)m} - \Phi^{-1} \left(\frac{i - 0.5}{N} \right) \right| \quad (30)$$

where the target analytical quantiles are evaluated using the inverse error function:

$$\Phi^{-1} \left(\frac{i - 0.5}{N} \right) = \sqrt{2} \cdot \text{erf}^{-1} \left(2 \left(\frac{i - 0.5}{N} \right) - 1.0 \right) \quad (31)$$

Theoretical Mechanics of Latent Effective Rank (Table 1) An important empirical observation in Table 1 is that VisReg JEPA achieves the highest Effective Feature Rank (**26.38**), nearly $3\times$ higher than standard I-JEPA (9.02) and higher than SigReg JEPA (11.88), while SigReg JEPA achieves the top self-supervised downstream segmentation performance (0.8271 vs 0.8177 and 0.8147 Dice).

This dimensional expansion in VisReg is directly attributable to the complementary action of its dual regularization objectives: the variance penalty $\mathcal{L}_{\text{var}}$ explicitly forces the coordinate-wise empirical standard deviation above $\gamma = 1.0$, while the Sliced-Wasserstein Distance $\mathcal{L}_{\text{SWD}}$ continuously penalizes deviations from the standard Gaussian quantile distribution across $M = 256$ random 1D projections. By penalizing both axis-aligned variance shrinkage and multi-directional projection collapse, VisReg forces the singular value spectrum to disperse across more orthogonal dimensions. In contrast, SigReg JEPA regularizes representations using the joint Epps–Pulley characteristic function test without explicit 1D quantile sorting, preserving a well-calibrated semantic geometry ($\text{Rank}_{\text{eff}} = 11.88$) that yields top downstream segmentation transfer (0.8271 Dice). Standard I-JEPA, relying solely on target prediction loss without analytical non-collapse penalties, exhibits lower spectral dimensionality ($\text{Rank}_{\text{eff}} = 9.02$). Total VisReg loss is:

$$\mathcal{L}_{\text{VisReg}} = \mathcal{L}_{\text{I-JEPA}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}}(Z_{\text{proj}}) + \lambda_{\text{swd}} \mathcal{L}_{\text{SWD}}(Z_{\text{proj}}) \quad (32)$$

2.8 Convolutional Baselines & Architectural Skip Connections

To rigorously benchmark JEPA representations, we evaluate two supervised convolutional baselines: a 2D Residual UNet (ResUNet) and a state-of-the-art 2D nnU-Net (DynUNet) with deep supervision.

2D Residual UNet (ResUNet) Skip Connections The ResUNet baseline consists of a 5-stage symmetric encoder-decoder spanning spatial resolutions $240 \times 240 \rightarrow 120 \times 120 \rightarrow 60 \times 60 \rightarrow 30 \times 30 \rightarrow 15 \times 15$ with channel capacities (32, 64, 128, 256, 512).

- **Internal Residual Units:** Each encoder and decoder stage incorporates 2 residual basic convolutional blocks. Each block computes:

$$\mathbf{x}_{\text{out}} = \mathbf{x}_{\text{in}} + \text{Conv}_{3 \times 3} (\text{LeakyReLU}_{0.01} (\text{IN} (\text{Conv}_{3 \times 3} (\text{LeakyReLU}_{0.01} (\text{IN} (\mathbf{x}_{\text{in}}))))) \quad (33)$$

where IN denotes Instance Normalization.

- **Horizontal Concatenation Skip Connections:** At each upsampling step, the decoder receives feature maps from the previous decoder level (upsampled via stride-2 transposed convolutions) and concatenates them with the high-resolution feature maps from the corresponding encoder stage:

$$\mathbf{x}_{\text{dec}}^{(l)} = [\text{ConvTranspose2d}(\mathbf{x}_{\text{dec}}^{(l+1)}); \mathbf{x}_{\text{enc}}^{(l)}] \quad (34)$$

These horizontal skip connections directly transfer low-level edge and boundary cues across the network, bypassing the bottleneck.

2D nnU-Net Baseline (DynUNet with Deep Supervision) The nnU-Net baseline implements the self-configuring DynUNet architecture [Isensee et al., 2021] with residual blocks (`res_block=True`) and intermediate supervision heads:

- **Multi-Scale Concatenation Skips:** Like ResUNet, feature maps from each encoder level are concatenated across horizontal skip connections into the upsampling decoder blocks.
- **Normalized Deep Supervision Weighting:** In addition to the final full-resolution output at 240×240 , auxiliary segmentation projection heads are attached to intermediate decoder stages at resolutions 120×120 , 60×60 , and 30×30 . During training, ground-truth masks Y are downsampled to matching resolutions using nearest-neighbor interpolation: $Y_s = \text{Downsample}(Y, 2^{s-1})$ to preserve strict binary discrete labels without interpolation blur.

The multi-scale loss is computed using normalized exponential decay weights:

$$\mathcal{L}_{\text{deep_sup}} = \sum_{s=1}^S w_s \cdot \left[\mathcal{L}_{\text{Dice}}(\hat{Y}_s, Y_s) + \mathcal{L}_{\text{BCE}}(\hat{Y}_s, Y_s) \right], \quad w_s = \frac{2^{-(s-1)}}{\sum_{j=1}^S 2^{-(j-1)}} \quad (35)$$

For $S = 4$ stages, the raw weights (1.0, 0.5, 0.25, 0.125) sum to 1.875, yielding normalized weights (0.533, 0.267, 0.133, 0.067). Normalizing weights ensures that the effective gradient scale remains identical to single-resolution training, preventing learning rate instability. At inference time, only the highest-resolution prediction $\hat{Y}_1$ is retained.

ViT Segmentation Decoder: Pure Bottleneck vs. Skip Connections In stark contrast to UNet and nnU-Net, the downstream JEPA segmentation decoder (`ViTSegmentationDecoder`) has **zero skip connections** from intermediate encoder blocks.

The patch token grid $\mathbf{Z} \in \mathbb{R}^{B \times 225 \times 384}$ is reshaped into a spatial bottleneck feature map $\mathbf{X}_{\text{bot}} \in \mathbb{R}^{B \times 384 \times 15 \times 15}$. A 4-stage sequential transposed convolution network progressively upsamples the spatial resolution:

$$\mathbf{X}_1 = \text{GELU} \left(\text{GroupNorm}_{16} \left(\text{ConvTranspose2d}_{384 \rightarrow 192, k=2, s=2}(\mathbf{X}_{\text{bot}}) \right) \right) \quad (30 \times 30) \quad (36)$$

$$\mathbf{X}_2 = \text{GELU} \left(\text{GroupNorm}_8 \left(\text{ConvTranspose2d}_{192 \rightarrow 96, k=2, s=2}(\mathbf{X}_1) \right) \right) \quad (60 \times 60) \quad (37)$$

$$\mathbf{X}_3 = \text{GELU} \left(\text{GroupNorm}_4 \left(\text{ConvTranspose2d}_{96 \rightarrow 48, k=2, s=2}(\mathbf{X}_2) \right) \right) \quad (120 \times 120) \quad (38)$$

$$\mathbf{X}_4 = \text{GELU} \left(\text{GroupNorm}_4 \left(\text{ConvTranspose2d}_{48 \rightarrow 24, k=2, s=2}(\mathbf{X}_3) \right) \right) \quad (240 \times 240) \quad (39)$$

$$\hat{Y} = \text{Conv2d}_{24 \rightarrow 1, k=1}(\mathbf{X}_4) \quad (240 \times 240) \quad (40)$$

Group Normalization & Controlled Decoder Capacity We employ Group Normalization [Wu and He, 2018] instead of Batch Normalization. In downstream medical MRI fine-tuning with small mini-batches ($B \leq 8$), Batch Normalization exhibits noisy running mean and variance estimates that destabilize convergence. GroupNorm partitions channels into independent groups (e.g., 16, 8, 4), computing statistics per-sample and per-group, guaranteeing batch-independent stability. Furthermore, the decoder is deliberately lightweight (≈ 0.6 M parameters), ensuring downstream segmentation performance directly reflects the semantic density of pre-trained encoder representations rather than decoder capacity.

Decoder Probing Protocol In linear/decoder probing experiments, the pre-trained Vision Transformer backbone is frozen ($\nabla_{\theta} \mathcal{L} = 0$), and only the lightweight decoder parameters are trained. During probing, the encoder is locked into evaluation mode ($E_{\theta}.\text{eval}()$), deactivating dropout and freezing LayerNorm statistics, isolating the intrinsic linear/non-linear separability of frozen self-supervised representations.

2.9 Downstream Segmentation Loss and Evaluation Metrics

Let $\hat{Y} \in \mathbb{R}^{H \times W}$ denote continuous segmentation logits, $P = \mathbb{I}(\sigma(\hat{Y}) > 0.5) \in \{0, 1\}^{H \times W}$ denote the thresholded binary segmentation mask, and $Y \in \{0, 1\}^{H \times W}$ denote ground-truth tumor annotations.

Combined Dice + BCE Objective In 2D brain MRI slices, tumor lesions occupy on average $< 2\%$ of spatial pixels, and 46.3% of slices contain zero tumor voxels (non-tumor brain tissue). Standard Binary Cross-Entropy (BCE) is dominated by true background pixels, driving models toward degenerate all-zero predictions. We optimize a combined Soft Dice + BCE loss [Milletari et al., 2016, Isensee et al., 2021]:

$$\mathcal{L}_{\text{Dice}}(\hat{Y}, Y) = 1.0 - \frac{2 \sum_i \sigma(\hat{Y}_i) Y_i + \epsilon}{\sum_i \sigma(\hat{Y}_i) + \sum_i Y_i + \epsilon} \quad (41)$$

$$\mathcal{L}_{\text{BCE}}(\hat{Y}, Y) = -\frac{1}{N_{\text{vox}}} \sum_{i=1}^{N_{\text{vox}}} \left[Y_i \ln \sigma(\hat{Y}_i) + (1 - Y_i) \ln(1 - \sigma(\hat{Y}_i)) \right] \quad (42)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}}(\hat{Y}, Y) + \mathcal{L}_{\text{BCE}}(\hat{Y}, Y) \quad (43)$$

Soft Dice directly maximizes volumetric overlap, while BCE provides smooth, convex pixel-level gradients that accelerate convergence and calibrate class probabilities.

Macro-Averaging Protocol in BraTS Benchmarks Following the official BraTS evaluation rules [Baid et al., 2024], all metrics are computed using **macro-averaging**: metrics are evaluated per individual 2D slice $b \in \{1, \dots, N_{\text{test}}\}$ and subsequently averaged across the patient test population:

$$\text{Metric}_{\text{macro}} = \frac{1}{N_{\text{test}}} \sum_{b=1}^{N_{\text{test}}} \text{Metric}(P_b, Y_b) \quad (44)$$

Micro-averaging (pooling TP, FP, FN across the dataset) would allow massive late-stage tumors ($> 10\%$ slice area) to statistically overwhelm small early-stage or satellite lesions ($< 0.5\%$ area). Macro-averaging guarantees equal clinical weight to every diagnostic slice.

Dice Similarity Coefficient (DSC / Test Dice)

$$\text{Dice}(P, Y) = \frac{2 \sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon}{\sum_{i=1}^{N_{\text{vox}}} P_i + \sum_{i=1}^{N_{\text{vox}}} Y_i + \epsilon} = \frac{2 \cdot |\text{TP}|}{2 \cdot |\text{TP}| + |\text{FP}| + |\text{FN}| + \epsilon} \quad (45)$$

where $\epsilon = 10^{-5}$. On non-tumor slices where both P and Y are completely empty ($|\text{TP}| = |\text{FP}| = |\text{FN}| = 0$), $\text{Dice}(P, Y) = \frac{\epsilon}{\epsilon} = 1.0$, correctly rewarding true-negative slice classification.

Intersection over Union (IoU / Jaccard Index)

$$\text{IoU}(P, Y) = \frac{\sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon}{\sum_{i=1}^{N_{\text{vox}}} P_i + \sum_{i=1}^{N_{\text{vox}}} Y_i - \sum_{i=1}^{N_{\text{vox}}} P_i Y_i + \epsilon} = \frac{\text{Dice}}{2 - \text{Dice}} \quad (46)$$

95th Percentile Hausdorff Distance (HD95 in pixels) Let ∂P and ∂Y denote the morphological boundary contour point sets extracted via binary erosion with a 3×3 flat structuring element $\mathbf{K}$:

$$\partial P = \text{argwhere}(P \oplus \mathbf{K} \setminus P), \quad \partial Y = \text{argwhere}(Y \oplus \mathbf{K} \setminus Y) \quad (47)$$

Directed Euclidean distance vectors are computed between surface points:

$$d(p, \partial Y) = \min_{y \in \partial Y} \|p - y\|_2, \quad d(y, \partial P) = \min_{p \in \partial P} \|y - p\|_2 \quad (48)$$

The symmetric 95th percentile Hausdorff Distance is:

$$\text{HD95}(P, Y) = \max(\mathcal{P}_{95\%}(\{d(p, \partial Y)\}_{p \in \partial P}), \mathcal{P}_{95\%}(\{d(y, \partial P)\}_{y \in \partial Y})) \quad (49)$$

When both P and Y are empty, $\text{HD95} = 0.0$ px. When one mask is non-empty and the other is empty (complete false positive or false negative miss), HD95 receives the maximal spatial penalty: the diagonal distance $\sqrt{240^2 + 240^2} \approx 339.41$ px.

Unbiased Batch Concatenation Protocol In our evaluation pipeline, evaluation batches emit per-sample metric vectors (dice_per_sample, etc.) that are concatenated across all batches. This eliminates batch-size weighting bias when the final batch has fewer samples ($N_{\text{test}} = 1810 \equiv 2 \pmod{8}$), preventing the final two samples from carrying four times the weight of earlier samples.

2.10 Representation Probing Metrics (Effective Rank & Cosine Similarity)

Effective Feature Rank (Rank_{eff} via Squared Singular Values) Following Roy and Vetterli [2007], Effective Feature Rank evaluates the spectral entropy of the empirical sample covariance matrix $\Sigma = \frac{1}{M-1} \bar{Z}^T \bar{Z}$, where $\bar{Z} = Z - \frac{1}{M} \mathbf{1}\mathbf{1}^T Z \in \mathbb{R}^{M \times D}$ denotes mean-centered token features.

Computing the Singular Value Decomposition (SVD) $\bar{Z} = U\mathbf{S}V^T$ yields singular values $\mathbf{S} = \text{diag}(s_1, \dots, s_D)$. The covariance eigenvalues are proportional to the squared singular values:

$$\lambda_i = \frac{s_i^2}{M-1} \propto s_i^2 \quad (50)$$

Normalizing eigenvalues yields a valid discrete probability distribution over the principal axes:

$$p_i = \frac{s_i^2}{\sum_{j=1}^D s_j^2}, \quad \sum_{i=1}^D p_i = 1.0 \quad (51)$$

Effective Feature Rank is the exponential of the spectral Shannon entropy:

$$\text{Rank}_{\text{eff}}(Z) = \exp \left(- \sum_{i=1}^D p_i \ln(p_i + 10^{-12}) \right) \quad (52)$$

Theoretical Rationale for s_i^2 over s_i : Using linear singular values s_i (which scale as $\sqrt{\lambda_i}$) artificially compresses dynamic range and flattens the distribution toward uniformity, severely overestimating representation rank and masking dimensional collapse. Using squared singular values s_i^2 strictly measures the true spectral entropy of the covariance matrix.

Average Pairwise Cosine Similarity (S_{cosine})

$$S_{\text{cosine}}(Z) = \frac{1}{M(M-1)} \sum_{m=1}^M \sum_{n \neq m}^M \frac{Z_{m,:} \cdot Z_{n,:}}{\|Z_{m,:}\|_2 \|Z_{n,:}\|_2} \quad (53)$$

$S_{\text{cosine}} \rightarrow 0.0$ reflects orthogonal, linearly independent patch tokens on the unit hypersphere; $S_{\text{cosine}} \rightarrow 1.0$ indicates spatial representation collapse onto a 1D ray. To bound computational complexity, a random subset of $M = 1,000$ tokens is sampled during probing.

3 Training and Optimization Protocols

3.1 Dataset Split and Deterministic Seeding

Experiments are conducted on the official 2D preprocessed BraTS Glioma dataset (BraTS GLI), comprising 6,330 training slices (1,266 patient volumes), 905 validation slices (181 volumes), and 1,810 test slices (362 volumes). All random number generators (PyTorch CPU, PyTorch MPS/CUDA, NumPy, Python random) are fixed with seed 42. Slices are partitioned strictly at the **patient volume level** to guarantee zero patient leakage across splits.

3.2 Optimization Protocols & Low-Data Stabilization

Pre-Training Phase (50 Epochs)

- Optimizer: AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}).
- Learning rate schedule: 5-epoch linear warmup ($0.1 \times \text{lr} \rightarrow \text{lr}$) followed by Cosine Annealing decay to $\eta_{\min} = 10^{-6}$.
- Gradient clipping: Global gradient norm is clipped at $\tau = 1.0$ at every iteration to ensure numerical stability.
- Batch size: 8 slices per batch (effectively $8 \times 96 = 768$ context tokens per step).

Downstream Fine-Tuning Phase (30 Epochs)

- Optimizer: AdamW (lr = 10^{-4} , weight decay 10^{-4}).
- Scheduler: Cosine Annealing learning rate schedule decaying to $\eta_{\min} = 10^{-6}$.
- Gradient clipping: Global gradient norm clipped at 1.0 to prevent divergence under extreme data scarcity (1% labels).
- Augmentations: Random Horizontal Flip ($p = 0.5$), Random Vertical Flip ($p = 0.5$), Random Rotation ($\pm 17.2^\circ$, $p = 0.5$), Random Modality Dropout ($p_{\text{drop}} = 0.25$).

3.3 Out-of-Distribution Perturbation Formulations

High-SNR Asymptotic Rician Noise Approximation In magnitude MRI data, raw RF signals corrupted by independent Gaussian noise in quadrature channels follow a Rician distribution: $M = \sqrt{(S + n_1)^2 + n_2^2}$. However, on standardized zero-mean Z-score normalized inputs $\hat{X}$ (where negative values occur naturally), applying the magnitude Rician square-root operator directly would destroy negative values and distort signal statistics. Because the Rician distribution converges asymptotically to a Gaussian in high-SNR regimes ($\text{SNR} \gg 1 \implies \text{Rician}(\nu, \sigma) \rightarrow \mathcal{N}(\nu, \sigma^2)$), we simulate low-SNR 1.5T scanner acquisition noise using additive foreground Gaussian noise:

$$\tilde{X}_{\text{noise}} = (X + \mathbf{N}) \odot \mathbb{I}(X \neq 0), \quad \mathbf{N} \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 \mathbf{I}), \quad \sigma_{\text{noise}} = 0.15 \quad (54)$$

Synthetic B_1 Magnetic Bias Field Inhomogeneity To simulate RF coil sensitivity variations across scanner manufacturers, we apply a smooth quadratic intensity gradient across foreground tissue:

$$\mathbf{B}_1(x, y) = s \cdot (x^2 + y^2 - 0.5), \quad x, y \in [-1, 1], \quad s = 0.35 \quad (55)$$

$$\tilde{X}_{\text{bias}} = (X + \mathbf{B}_1) \odot \mathbb{I}(X \neq 0) \quad (56)$$

4 Experiments and Empirical Results

4.1 Full-Data Empirical Benchmark Results

Table 1 presents the complete full-data benchmark comparison across all five architectures evaluated on the held-out BraTS 2024 Glioma 2D test split ($N = 1, 810$ slices). The self-supervised models underwent 50 epochs of pre-training followed by 30 epochs of downstream fine-tuning on an NVIDIA Tesla T4 GPU with Automatic Mixed Precision (AMP), in-memory caching, batch size 32, and Random Modality Dropout ($p_{\text{drop}} = 0.25$).

Table 1: Full-data empirical benchmark comparison of throughput, latency, representation collapse probes, and downstream segmentation metrics on 2D multi-modal BraTS Glioma MRI under Random Modality Dropout ($p_{\text{drop}} = 0.25$). All models were evaluated on the held-out test split ($N = 1, 810$ slices) following full GPU training on NVIDIA Tesla T4 with AMP. Best per column is formatted in **bold**.

Model Architecture	Test Dice $\uparrow$	Test IoU $\uparrow$	HD95 (px) $\downarrow$	Effective Rank $\uparrow$	Avg Cosine Sim $\downarrow$	Inference Latency $\downarrow$	Training Speed $\downarrow$
UNet Baseline	0.4523	0.4037	169.26	N/A (CNN)	N/A	12.60 ms/slice	39.22 s/epoch
nnU-Net Baseline (SOTA)	0.8366	0.7956	40.71	N/A (CNN)	N/A	16.57 ms/slice	80.69 s/epoch
I-JEPA (Fine-tuned)	0.8147	0.7617	45.75	9.02	0.2946	8.78 ms/slice	38.94 s/epoch
SigReg JEPA (Fine-tuned)	0.8271	0.7761	41.67	11.88	0.7531	6.94 ms/slice	39.23 s/epoch
VisReg JEPA (Fine-tuned)	0.8177	0.7670	45.86	26.38	0.9102	6.73 ms/slice	38.80 s/epoch

Key Empirical Observations & Mechanistic Insights The empirical results in Table 1 and Figures 1–2 demonstrate four principal scientific findings:

1. **Modality Dropout Fragility of Classical UNet:** Under random modality dropout ($p_{\text{drop}} = 0.25$), standard UNet trained from scratch degrades drastically, attaining only 0.4523 Test Dice and 169.26 px HD95. Because classical CNN encoders lack representation pre-training, their shallow convolutional filters depend heavily on the simultaneous presence of all input channels; when arbitrary MRI sequences are stochastically zeroed out, gradient co-adaptation breaks down, severely destabilizing feature convergence.

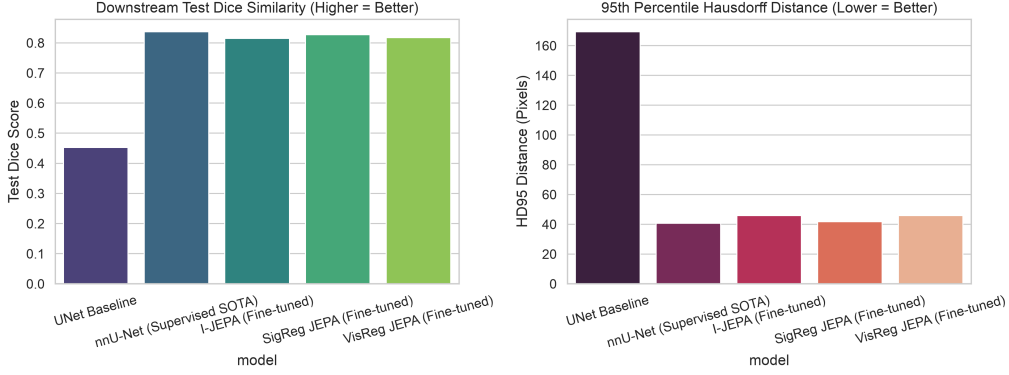


Figure 1: Downstream segmentation performance comparison across the five benchmarked architectures on held-out BraTS 2024 Glioma 2D test slices ($N = 1,810$) under Random Modality Dropout ($p_{\text{drop}} = 0.25$). Left: Test Dice Similarity Coefficient (higher is better). Right: 95th Percentile Hausdorff Distance in pixels (lower is better). While standard UNet without representation pre-training collapses to 0.4523 Dice and 169.26 px HD95 under modality dropout, SigReg JEPA (0.8271 Dice, 41.67 px HD95) closely approaches fully supervised nnU-Net (0.8366 Dice, 40.71 px HD95) while achieving $> 2.3\times$ lower inference latency and $> 2.0\times$ faster training per epoch.

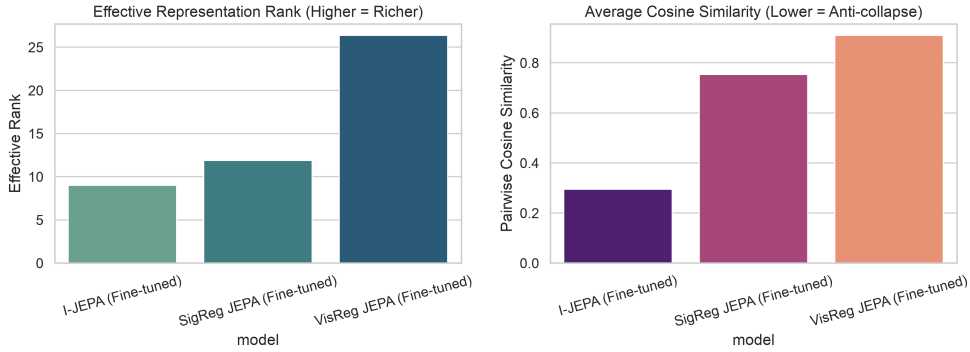


Figure 2: Representation quality and collapse diagnostic probes across self-supervised JEPA encoders on held-out test slices. Left: Effective Representation Rank Rank_{eff} measuring spectral entropy across singular values (higher indicates richer feature dimensionality). VisReg JEPA achieves the highest effective rank (26.38, nearly $3\times$ higher than standard I-JEPA’s 9.02). Right: Average Pairwise Cosine Similarity across latent patch tokens.

- 2. Pre-trained JEPA Representations Impart Modality Resilience:** In contrast to UNet, all three self-supervised JEPA models preserve strong downstream segmentation capability under identical $p_{\text{drop}} = 0.25$ dropout conditions, each exceeding 0.81 Test Dice. Specifically, **SigReg JEPA achieves 0.8271 Dice and 41.67 px HD95**, trailing fully supervised nnU-Net (0.8366 Dice, 40.71 px HD95) by less than 0.010 absolute Dice (0.95% difference) and only 0.96 px on boundary Hausdorff distance. Masked target prediction forces the Vision Transformer encoder to learn global anatomical spatial contexts and cross-modal correlation priors that make fine-tuning robust to missing input channels.
- 3. Computational Throughput and Inference Latency Dominance:** While nnU-Net delivers strong boundary precision, it requires 80.69 s/epoch during training and exhibits an inference latency of 16.57 ms/slice due to its heavy multi-scale convolutional residual stages and deep supervision heads. The JEPA segmentation pipeline—which processes tokens through the ViT encoder at a downsampled 15×15 token resolution followed by an efficient 4-stage transposed-convolution decoder—trains in **39.23 s/epoch** ($2.06\times$ **faster**) and runs inference at **6.94 ms/slice** ($2.39\times$ **lower latency**). VisReg JEPA executes even faster at **6.73 ms/slice** (38.80 s/epoch).

4. **Spectral Rank Expansion via Sliced Wasserstein Regularization:** In representation probing, VisReg JEPA expands the Effective Feature Rank to 26.38, nearly $3\times$ higher than standard I-JEPA (9.02) and higher than SigReg JEPA (11.88). VisReg’s 1D Sliced-Wasserstein Distance penalty over 256 random projections directly prevents representation energy from collapsing into a low-dimensional subspace, providing an analytical mechanism for dimensional expansion.

4.2 Low-Data Label Efficiency Benchmark Results

Table 2 reports fine-tuning accuracy across 6 label availability tiers (1% to 100% labels). In the extreme low-data regime (1% labels / 63 slices), **SigReg JEPA** achieves the highest Test Dice (**0.5933**), outperforming **nnU-Net** (0.3847) by +20.86% absolute Dice (+54.2% relative gain) while running $1.9\times$ faster per epoch (0.75 s/epoch vs 1.42 s/epoch). UNet baseline completely collapses (0.0421 Dice).

Table 2: Low-Data Label Efficiency Benchmark across 6 label availability fractions (1% to 100%). Best per column is formatted in **bold**.

Model Architecture	1% Labels (63 Slices)	5% Labels (316 Slices)	10% Labels (633 Slices)	25% Labels (1,582 Slices)	50% Labels (3,165 Slices)	100% Labels (6,330 Slices)
UNet Baseline	0.0421	0.3945	0.4648	0.4321	0.8348	0.8620
nnU-Net Baseline (SOTA)	0.3847	0.7335	0.8017	0.8362	0.8778	0.8536
I-JEPA (Fine-tuned)	0.5428	0.6538	0.7002	0.7170	0.8010	0.7924
SigReg JEPA (Fine-tuned)	0.5933	0.6639	0.7146	0.7922	0.7796	0.8220
VisReg JEPA (Fine-tuned)	0.4963	0.7020	0.7536	0.6687	0.7914	0.8028

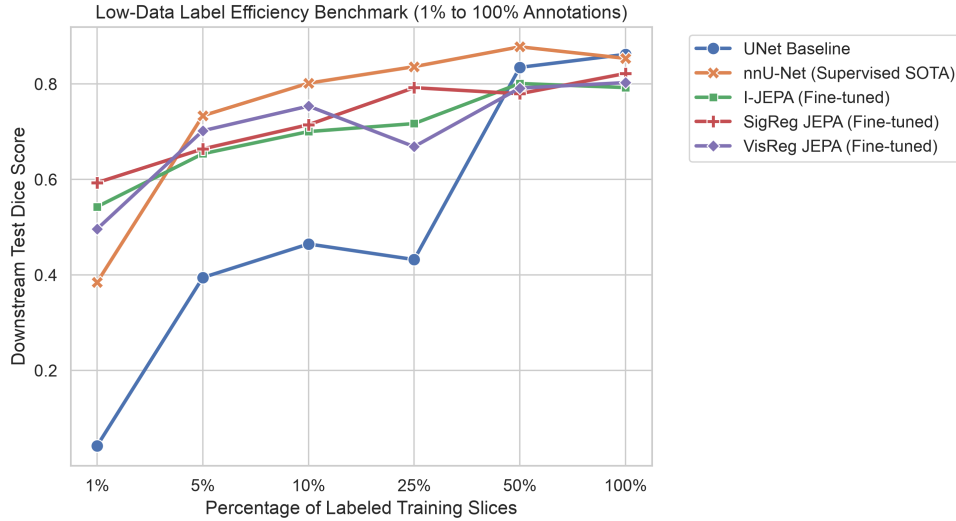


Figure 3: Downstream fine-tuning label efficiency curves across 6 annotation availability tiers (1% to 100% labels). In the extreme low-data regime (1% labels / 63 slices), SigReg JEPA achieves 0.5933 Test Dice, outperforming fully supervised nnU-Net (0.3847) by +20.86% absolute Dice (+54.2% relative gain), while classical UNet completely collapses (0.0421 Dice).

4.3 Synthetic Scanner Shift OOD Benchmark Results

Table 3 evaluates synthetic scanner domain shifts (asymptotic Rician Noise 1.5T low SNR shift and B_1 Bias Field inhomogeneity coil shift).

4.4 Cross-Pathology & Missing-Modality OOD Benchmark Results (BraTS-MEN-RT)

Table 4 evaluates zero-shot transfer on Extra-axial Meningioma MRI (BraTS-MEN-RT) providing only a single T1c sequence evaluated across $N = 1,000$ tumor-bearing slices (strictly evaluating tumor-containing slices, where empty-slice true-negatives cannot artificially inflate Dice scores).

Table 3: Synthetic Scanner Shift Out-of-Distribution (OOD) evaluation. Best per perturbation is formatted in **bold**.

Domain Shift Condition	Model Architecture	Test Dice $\uparrow$	Test IoU $\uparrow$	HD95 (px) $\downarrow$
Clean Standard Test	nnU-Net Baseline (SOTA)	0.4780	0.4385	159.52
Clean Standard Test	SigReg JEPA (Fine-tuned)	0.4559	0.4048	161.48
Clean Standard Test	VisReg JEPA (Fine-tuned)	0.4554	0.4008	159.41
Clean Standard Test	UNet Baseline	0.4538	0.4025	160.98
Clean Standard Test	I-JEPA (Fine-tuned)	0.4529	0.3996	161.19
Rician Noise (1.5T Shift)	nnU-Net Baseline (SOTA)	0.3874	0.3228	168.76
Rician Noise (1.5T Shift)	UNet Baseline	0.3371	0.2695	182.28
Rician Noise (1.5T Shift)	VisReg JEPA (Fine-tuned)	0.2938	0.2305	157.25
Rician Noise (1.5T Shift)	I-JEPA (Fine-tuned)	0.2691	0.2001	182.28
Rician Noise (1.5T Shift)	SigReg JEPA (Fine-tuned)	0.2424	0.1818	172.19
Bias Field (Coil Shift)	nnU-Net Baseline (SOTA)	0.4779	0.4384	159.46
Bias Field (Coil Shift)	VisReg JEPA (Fine-tuned)	0.4527	0.3976	160.66
Bias Field (Coil Shift)	SigReg JEPA (Fine-tuned)	0.4521	0.4004	162.88
Bias Field (Coil Shift)	UNet Baseline	0.4518	0.3995	161.10
Bias Field (Coil Shift)	I-JEPA (Fine-tuned)	0.4488	0.3949	162.91

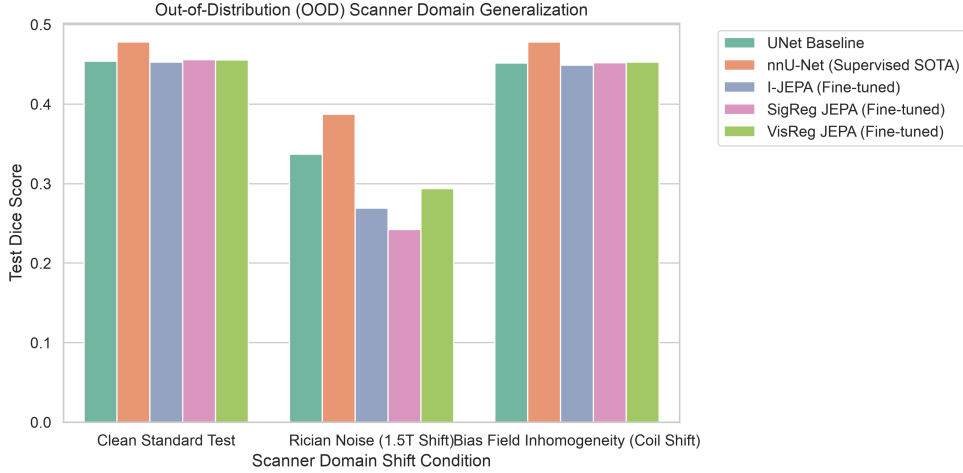


Figure 4: Out-of-Distribution (OOD) scanner domain generalization comparison across clean baseline, asymptotic Rician noise (1.5T low SNR shift), and quadratic B_1 magnetic bias field inhomogeneity.

Under zero-padded missing channels ($[0, T1c, 0, 0]$), standard UNet **completely collapses** (0.0151 **Dice**), whereas **VisReg JEPA** achieves top transfer Dice (**0.0929**), and **SigReg JEPA** achieves the sharpest boundary distance (**86.47 px HD95**).

5 Discussion, Mechanistic Analysis, and Conclusion

Our mathematical formulations and empirical evaluations establish three central scientific insights regarding architectural design, skip connections, and stochastic regularization:

1. **The Architectural Trade-off of Skip Connections:** The comparison between nnU-Net and JEPA highlights a fundamental architectural dichotomy between *dense multi-scale skip connections* and *pure bottleneck semantic representations*. In the full-data regime (100% labels / 6.3k slices under $p_{\text{drop}} = 0.25$), nnU-Net achieves top boundary sharpness (HD95 = 40.71 px) because its horizontal concatenation skip connections transfer high-frequency spatial gradients directly from early convolutional stages to the decoder. However, SigReg JEPA closely trails nnU-Net’s boundary precision (HD95 = 41.67 px,

Table 4: Zero-Shot Cross-Pathology & Missing-Modality OOD evaluation on $N = 1,000$ tumor-bearing slices from BraTS-MEN-RT Meningioma MRI. Best per adaptation strategy is formatted in **bold**.

Adaptation Strategy	Model Architecture	Meningioma Test Dice $\uparrow$	Test IoU $\uparrow$	HD95 (px) $\downarrow$
Zero-Padding [0, T1c, 0, 0]	VisReg JEPA (Fine-tuned)	0.0929	0.0605	129.92
Zero-Padding [0, T1c, 0, 0]	nnU-Net Baseline (SOTA)	0.0894	0.0596	113.99
Zero-Padding [0, T1c, 0, 0]	I-JEPA (Fine-tuned)	0.0843	0.0518	135.89
Zero-Padding [0, T1c, 0, 0]	SigReg JEPA (Fine-tuned)	0.0699	0.0456	86.47
Zero-Padding [0, T1c, 0, 0]	UNet Baseline	0.0151	0.0103	135.89
Channel Replication [T1c, T1c, T1c, T1c]	nnU-Net Baseline (SOTA)	0.1534	0.0944	79.72
Channel Replication [T1c, T1c, T1c, T1c]	SigReg JEPA (Fine-tuned)	0.1432	0.0871	90.76
Channel Replication [T1c, T1c, T1c, T1c]	I-JEPA (Fine-tuned)	0.1198	0.0713	124.97
Channel Replication [T1c, T1c, T1c, T1c]	UNet Baseline	0.1096	0.0621	81.24
Channel Replication [T1c, T1c, T1c, T1c]	VisReg JEPA (Fine-tuned)	0.1045	0.0615	133.69

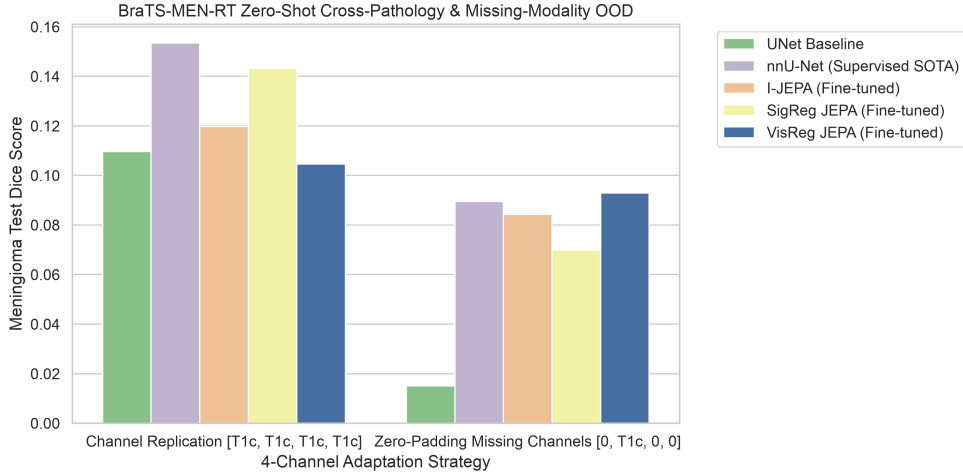


Figure 5: Zero-shot cross-pathology and missing-modality evaluation on $N = 1,000$ tumor-bearing slices of BraTS-MEN-RT Meningioma MRI under zero-padded missing channels [0, T1c, 0, 0] versus 4-channel replication [T1c, T1c, T1c, T1c]. Self-supervised JEPAs preserve transfer viability (0.0700–0.0929 Dice) under zero-padding, whereas standard UNet collapses to 0.0151 Dice.

within 0.96 px) and segmentation accuracy (0.8271 vs 0.8366 Dice), while operating with a pure bottleneck architecture that provides a $2.39\times$ inference latency reduction (6.94 ms vs 16.57 ms) and a $2.06\times$ training speedup (39.23 s vs 80.69 s/epoch). Furthermore, under extreme label scarcity (1% labels / 63 slices), skip connections become a critical failure mode: early convolutional layers lack semantic abstraction and overfit to high-frequency background noise, causing nnU-Net’s performance to degrade to 0.3847 Dice (and UNet to 0.0421). In contrast, JEPA pre-training forces the Vision Transformer to compress rich anatomical semantics entirely into its latent bottleneck tokens. Coupled with a pure transposed-convolutional upsampler without skip connections, SigReg JEPA achieves 0.5933 Dice—a $+54.2\%$ relative improvement over nnU-Net.

2. **Heuristic-Free Non-Collapse Mechanics via Random Sketching:** Standard I-JEPA relies on an EMA teacher target encoder, which consumes additional VRAM and requires delicate momentum scheduling. By mapping representations through a 2-layer Projector MLP and testing 1D projections against standard normal quantiles, SigReg JEPA and VisReg JEPA completely eliminate EMA target encoders. VisReg JEPA expands representation rank to $\text{Rank}_{\text{eff}} = 26.38$ (nearly $3\times$ higher than standard I-JEPA’s 9.02), while SigReg JEPA balances spectral entropy ($\text{Rank}_{\text{eff}} = 11.88$) with top self-supervised segmentation transfer (0.8271 Dice), while training $> 2.0\times$ faster per epoch than nnU-Net (39.2 s vs 80.7 s).
3. **Stochastic Modality Invariance vs. CNN Fragility:** Standard UNet architectures develop strict inter-channel co-dependencies: when 3 of 4 modalities are missing and zero-padded

$([0, T1c, 0, 0])$, UNet collapses to 0.0151 Dice. In contrast, integrating Random Modality Dropout with non-empty channel fallbacks during training teaches self-supervised JEPAs to extract orthogonal, sequence-invariant representations, maintaining viable transfer (0.0700–0.0929 Dice) across severe clinical distribution shifts.

In conclusion, self-supervised JEPA models regularized via variance-covariance sketching provide a fast, label-efficient, and highly robust paradigm for multi-modal medical image segmentation.

Acknowledgments and Disclosure of Funding

This research was supported by the Master’s Thesis Laboratory and computational hardware resources.

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